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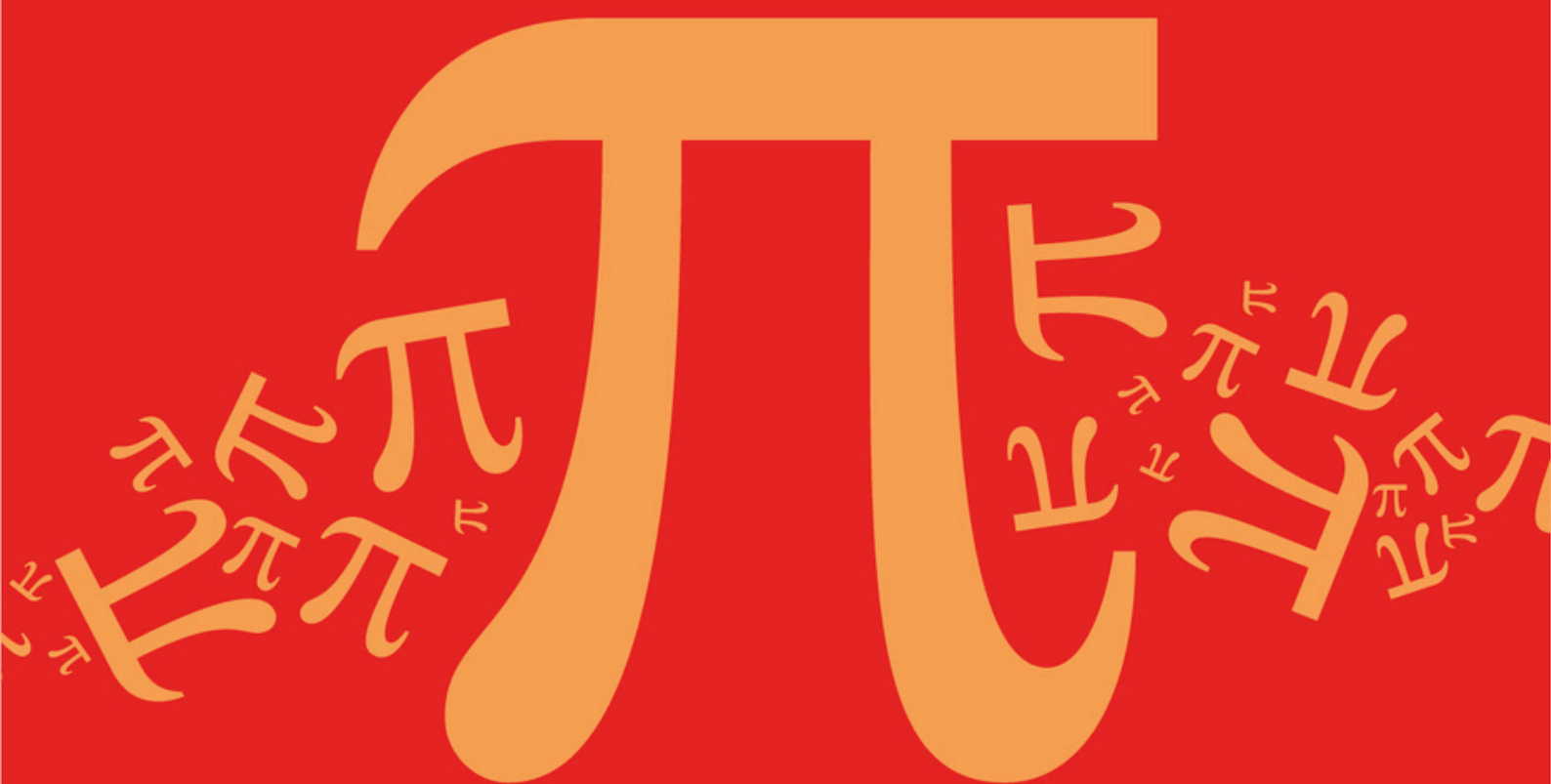
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AQA A Level Further Mathematics (7367)



Topic

DC: Network Flows

Sub topic

DC1: Interpret Directed Networks

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

DC: Network Flows

Sub topic

DC1: Interpret Directed Networks

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
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Name : _____

Date : _____

Time : _____

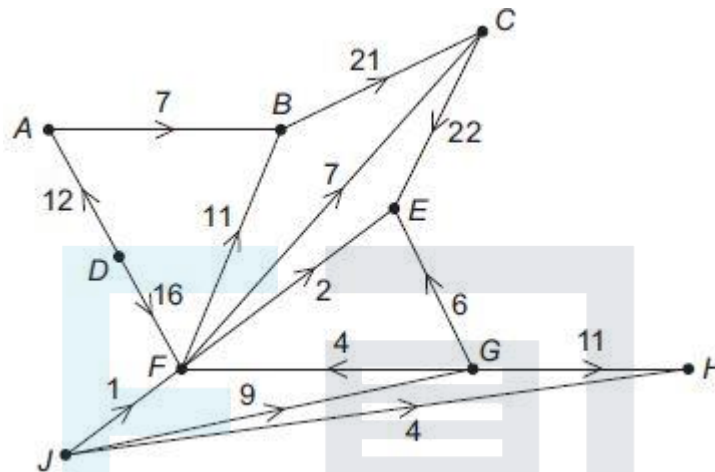
Total Marks Available : _____

Total Marks Received : _____

Q1.

The diagram shows a network of pipes.

Each pipe is labelled with its upper capacity in $\text{cm}^3 \text{ s}^{-1}$



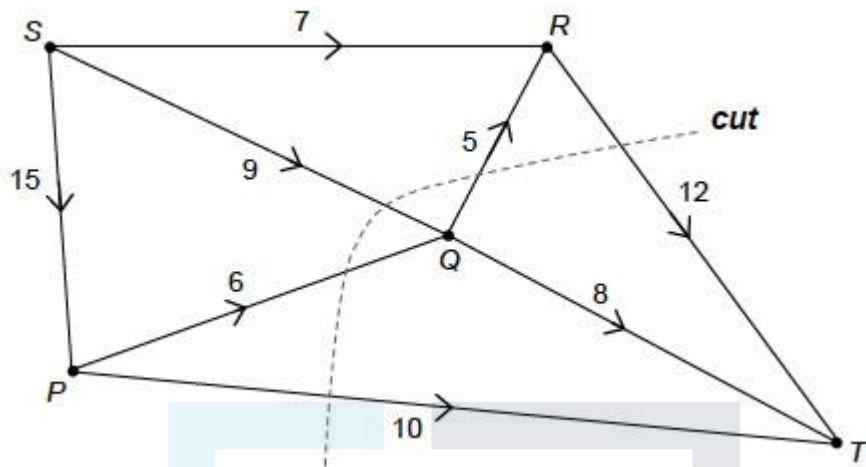
- (a) (i) Find the value of the cut given by $\{A, B, C, D, F, J\} \{E, G, H\}$. (1)
- (ii) State what can be deduced about the maximum flow through the network. (1)
- (b) (i) List the nodes which are sources of the network. (1)
- (ii) Add a supersource S to the network. (1)
- (c) (i) List the nodes which are sinks of the network. (1)
- (ii) Add a supersink T to the network. (1)
- (Total 6 marks)**

Q2.

The network shows a system of pipes, where S is the source and T is the sink.

The capacity, in litres per second, of each pipe is shown on each arc.

The cut shown in the diagram can be represented as $\{S, P, R\}, \{Q, T\}$.



- (a) Complete the table below to give the value of each of the 8 possible cuts.

Cut		Value
$\{S\}$	$\{P, Q, R, T\}$	31
$\{S, P\}$	$\{Q, R, T\}$	32
$\{S, Q\}$	$\{P, R, T\}$	
$\{S, R\}$	$\{P, Q, T\}$	
$\{S, P, Q\}$	$\{R, T\}$	30
$\{S, P, R\}$	$\{Q, T\}$	37
$\{S, Q, R\}$	$\{P, T\}$	35
$\{S, P, Q, R\}$	$\{T\}$	30

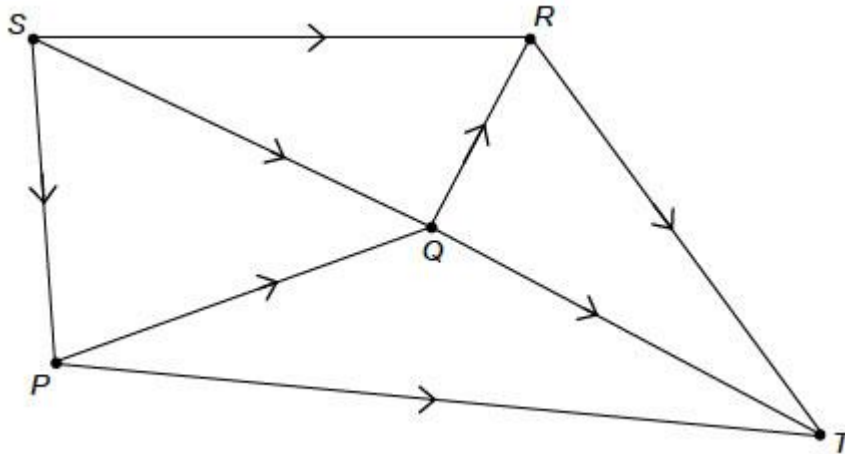
(1)

- (b) State the value of the maximum flow through the network.

Give a reason for your answer.

(1)

- (c) Indicate on the diagram a possible flow along each arc, corresponding to the maximum flow through the network.

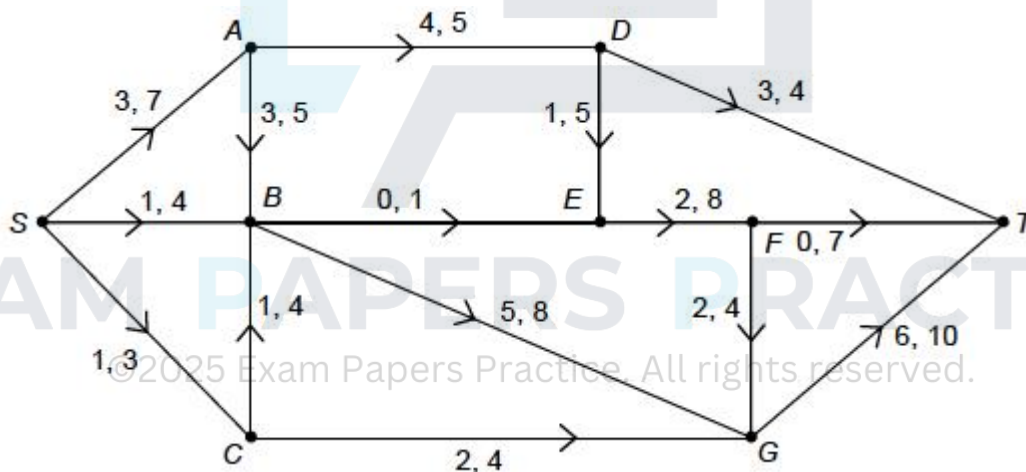


(2)
(Total 4 marks)

Q3.

The network shows a system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each arc.



- (a) There is a feasible flow from S to T .
- (i) Explain why arc AD must be at its lower capacity. (1)
- (ii) Explain why arc BE must be at its upper capacity. (1)
- (b) Explain why a flow of 11 litres per second through the network is impossible. (1)
- (c) The network in **Figure 1** shows a second system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each edge.

Figure 1

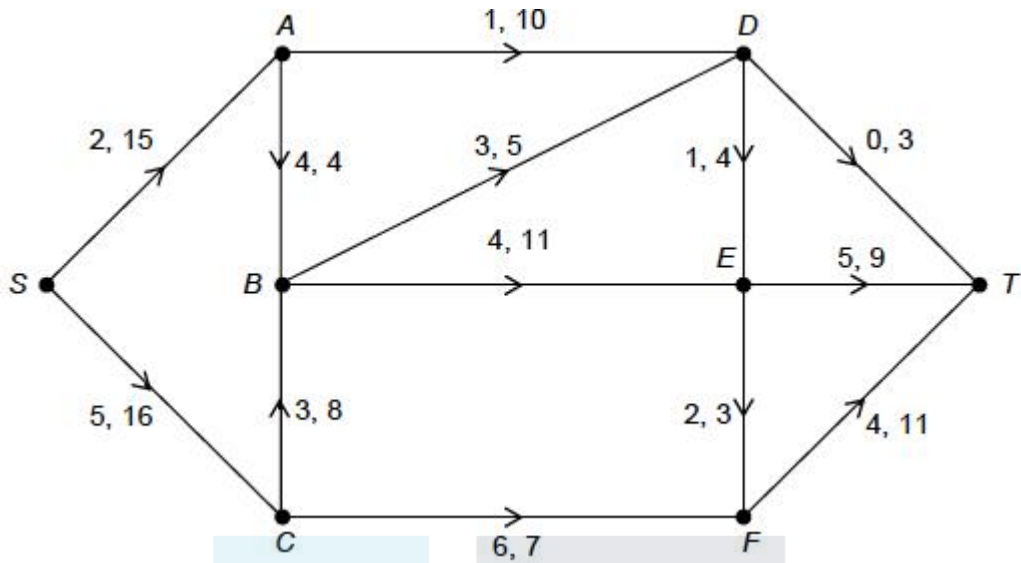
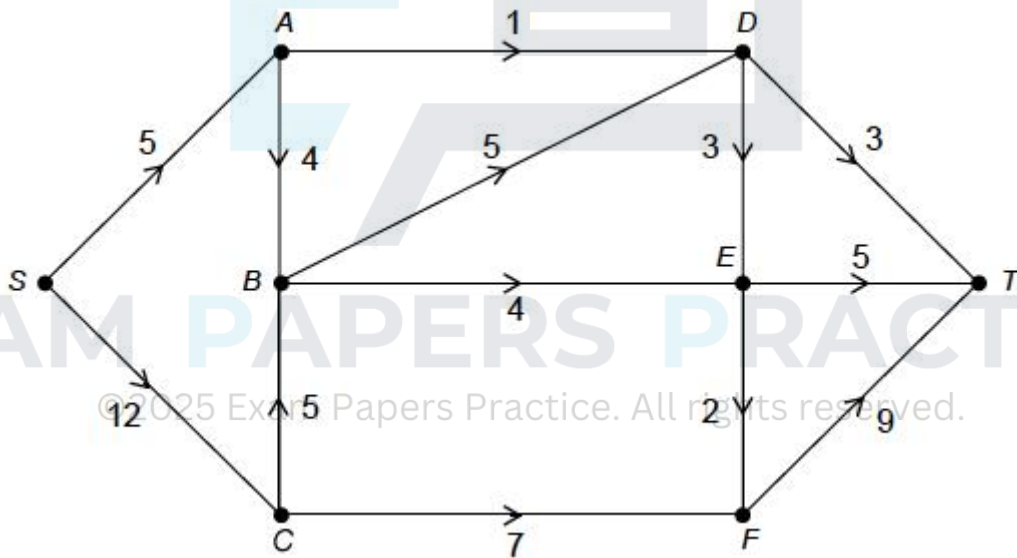


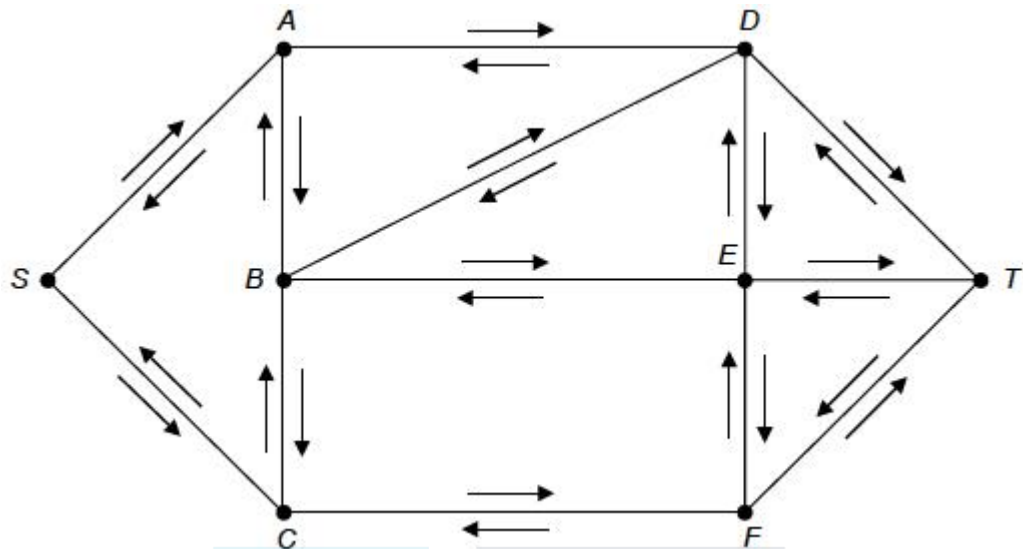
Figure 2 shows a feasible flow of 17 litres per second through the system of pipes.

Figure 2



- (i) Using Figures 1 and 2, indicate on Figure 3 potential increases and decreases in the flow along each arc.

Figure 3



(2)

- (ii) Use flow augmentation on **Figure 3** to find the maximum flow from S to T .

You should indicate any flow augmenting paths clearly in the table below and modify the potential increases and decreases of the flow on **Figure 3**.

(1)

Augmenting Path	Flow

- (iii) Prove the flow found in part (c)(ii) is maximum.

(1)

- (iv) Due to maintenance work, the flow through node E is restricted to 9 litres per second.

Interpret the impact of this restriction on the maximum flow through the system of pipes.

(2)

(Total 11 marks)

Q4.

- (a) When investigating three network flow problems, a student finds:

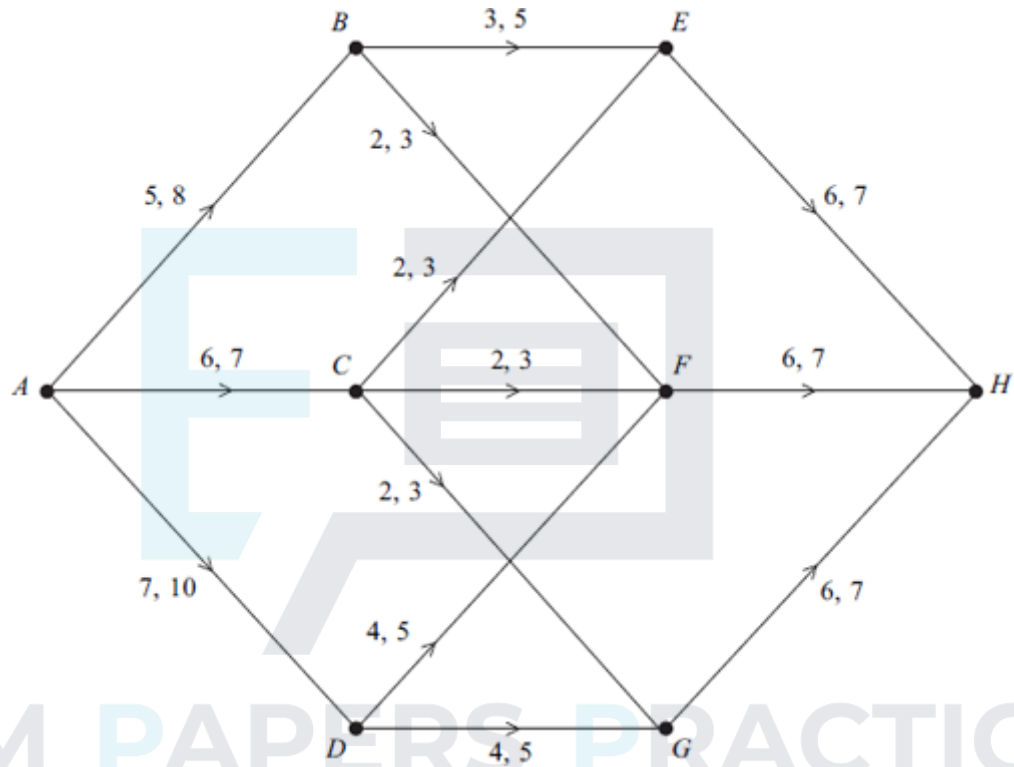
- (i) a flow of 50 and a cut with capacity 50;
- (ii) a flow of 35 and a cut with capacity 50;
- (iii) a flow of 50 and a cut with capacity 35.

In each case, write down what the student can deduce about the maximum flow.

(4)

- (b) The diagram below shows a network. The numbers on the arcs represent the minimum and maximum flow along each arc respectively.

By considering the flow at an appropriate vertex, explain why a flow is not possible through this network.



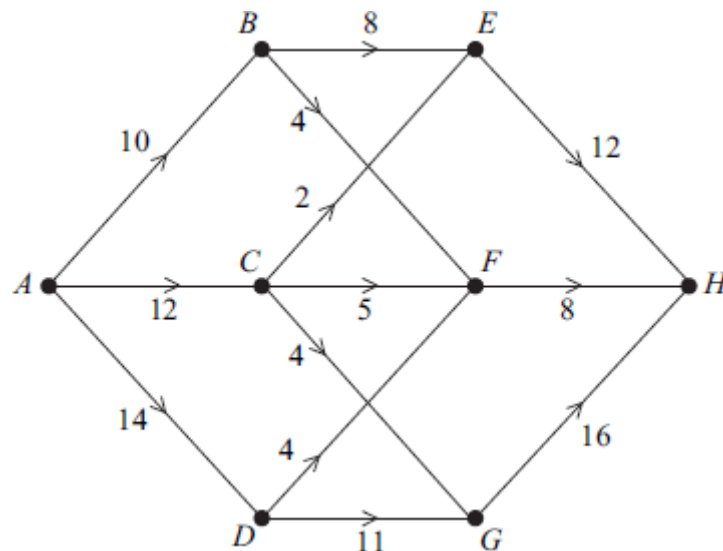
(2)

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(Total 6 marks)

Q5.

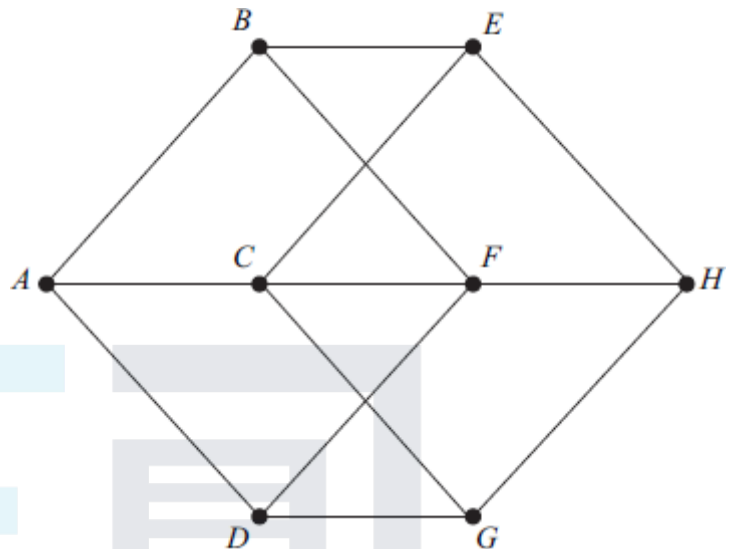
The network below represents a system of pipes. The capacity of each pipe, in litres per second, is indicated on the corresponding edge.



- (a) Find the maximum flow along each of the routes $ABEH$, $ACFH$ and $ADGH$ and enter their values in the table on **Figure 1**.

Figure 1

Route	Flow
$ABEH$	
$ACFH$	
$ADGH$	



(1)

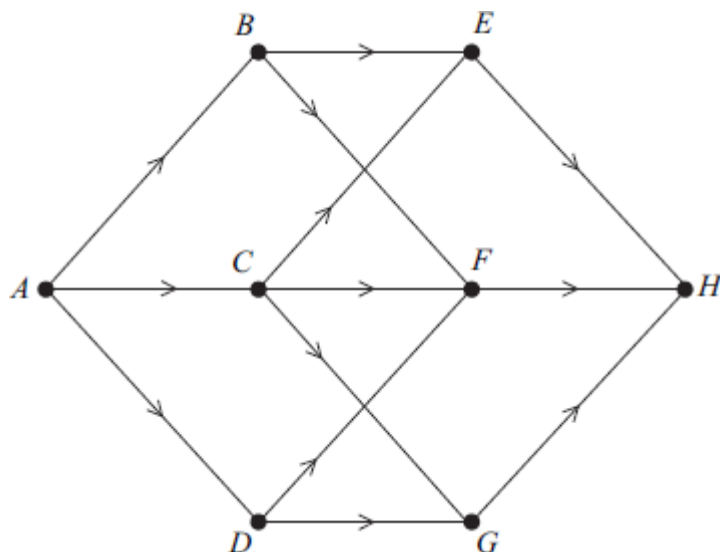
- (b) (i) Taking your answers to part (a) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow-augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (ii) State the value of the maximum flow and, on **Figure 2** below, illustrate a possible flow along each edge corresponding to this maximum flow.

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Figure 2



(2)

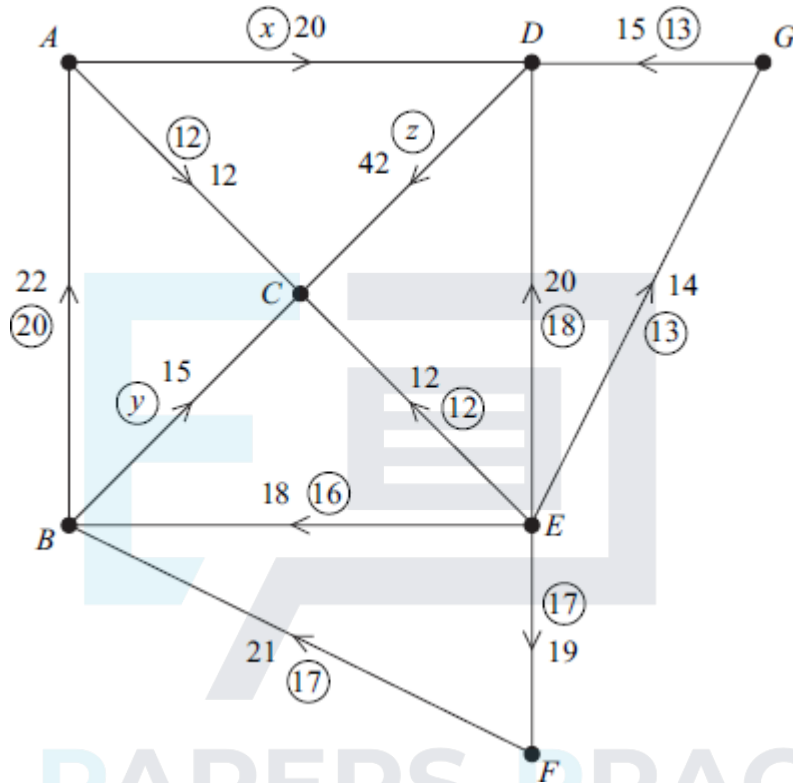
- (c) Confirm that you have a maximum flow by finding a cut of the same value. List the edges of your cut.

(1)

(Total 9 marks)

Q6.

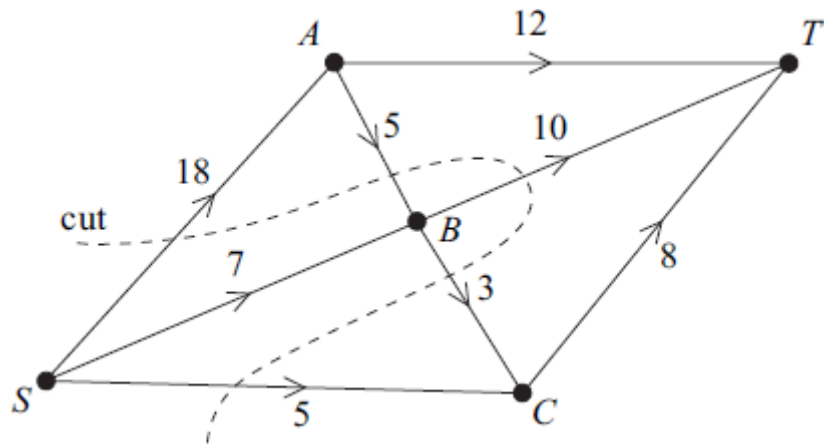
The network below represents a system of pipes. The number **not** circled on each edge represents the capacity of each pipe in litres per second. The number or letter in each circle represents an initial flow in litres per second.



- (a) Write down the capacity of edge EF . (1)
 - (b) State the source vertex. (1)
 - (c) State the sink vertex. (1)
 - (d) Find the values of x , y and z . (3)
 - (e) Find the value of the initial flow. (1)
 - (f) Find the value of a cut through the edges EB , EC , ED , EF and EG . (1)
- (Total 8 marks)

Q7.

- (a) The network shows a flow from S to T along a system of pipes, with the capacity in litres per second indicated on each edge.



- (i) Show that the value of the cut shown on the diagram is 36 .

(1)

- (ii) The cut shown on the diagram can be represented as $\{S, B\}, \{A, C, T\}$.

Complete the table below to give the value of each of the 8 possible cuts.

Cut		Value
$\{S\}$	$\{A, B, C, T\}$	30
$\{S, A\}$	$\{B, C, T\}$	29
$\{S, B\}$	$\{A, C, T\}$	36
$\{S, C\}$	$\{A, B, T\}$	33
$\{S, A, B\}$	$\{C, T\}$	
$\{S, A, C\}$	$\{B, T\}$	
$\{S, B, C\}$	$\{A, T\}$	
$\{S, A, B, C\}$	$\{T\}$	30

(3)

- (iii) State the value of the maximum flow through the network, giving a reason for your answer.

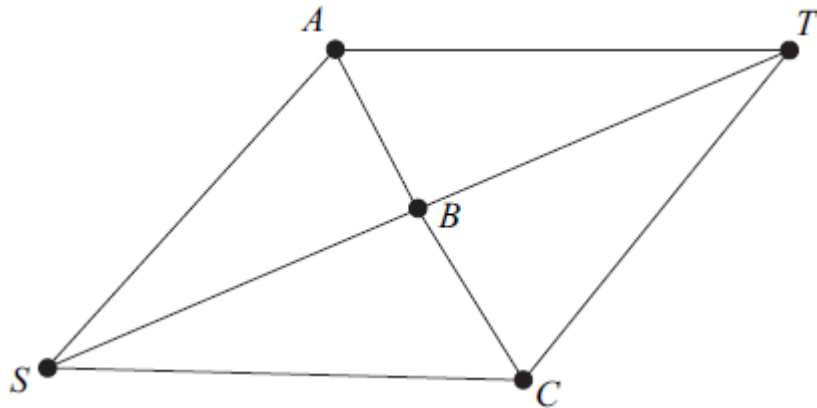
Maximum flow is _____

because _____

(2)

- (iv) Indicate on the diagram below a possible flow along each edge corresponding

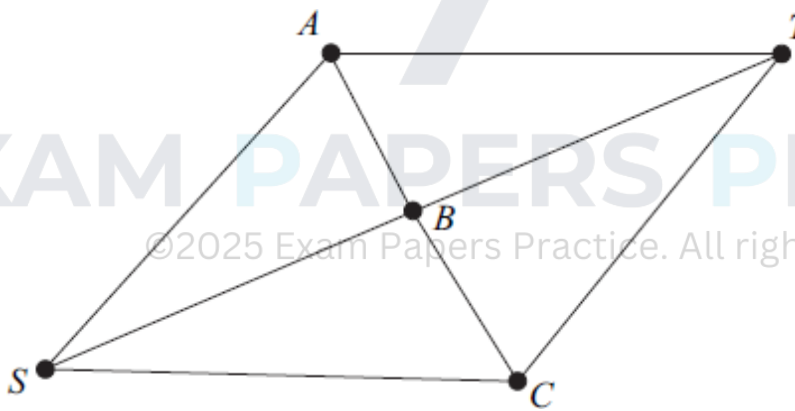
to this maximum flow.



(1)

(b) The capacities along SC and along AT are each increased by 4 litres per second.

- (i) Using your **values from part (a)(iv) as the initial flow**, indicate potential increases and decreases on the diagram below and use the labelling procedure to find the new maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the diagram.

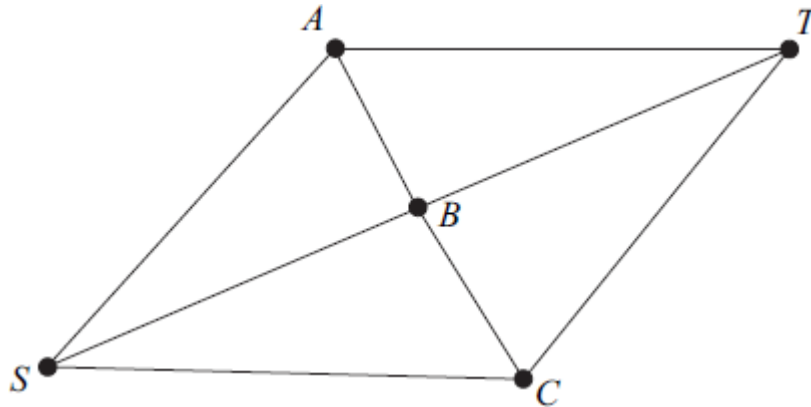


Path	Additional Flow

(6)

- (ii) Use your results from part (b)(i) to illustrate the flow along each edge that gives this new maximum flow, and state the value of the new maximum flow.

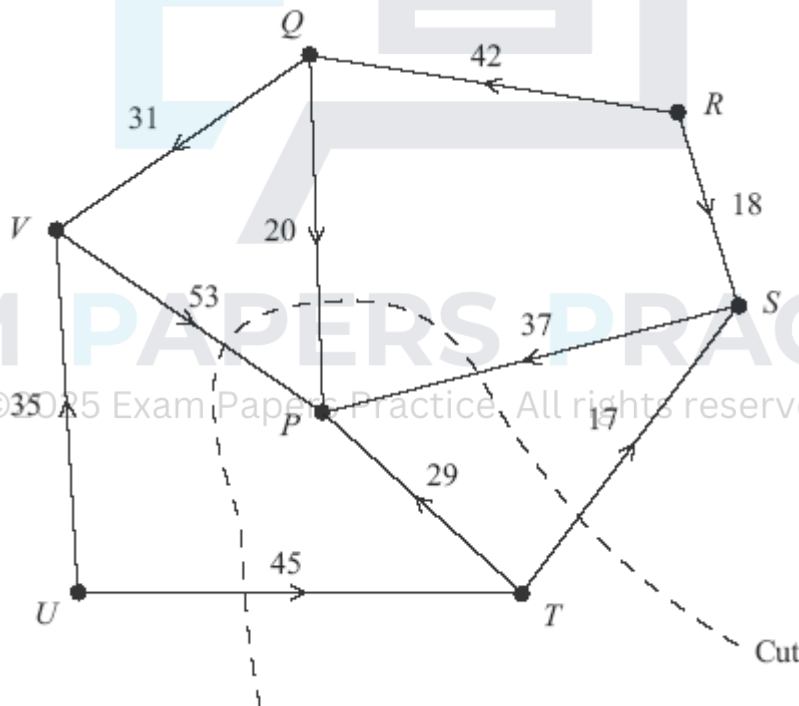
New maximum flow is _____



(3)
(Total 16 marks)

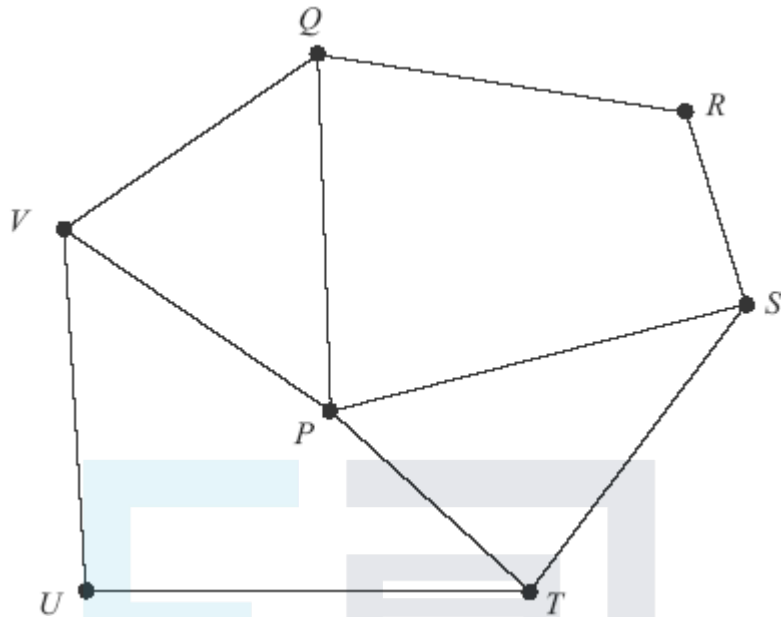
Q8.

The network shows the routes along corridors from two arrival gates to the passport control area, P , in a small airport. The number on each edge represents the maximum number of passengers that can travel along a particular corridor in one minute.



- (a) State the vertices that represent the arrival gates. (1)
- (b) Find the value of the cut shown on the network. (1)
- (c) State the maximum flow along each of the routes $UTSP$ and $RQVP$. (2)
- (d) (i) Taking your answers to part (c) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

Figure 1



Route	Value of Flow
<i>UTSP</i>	
<i>RQVP</i>	

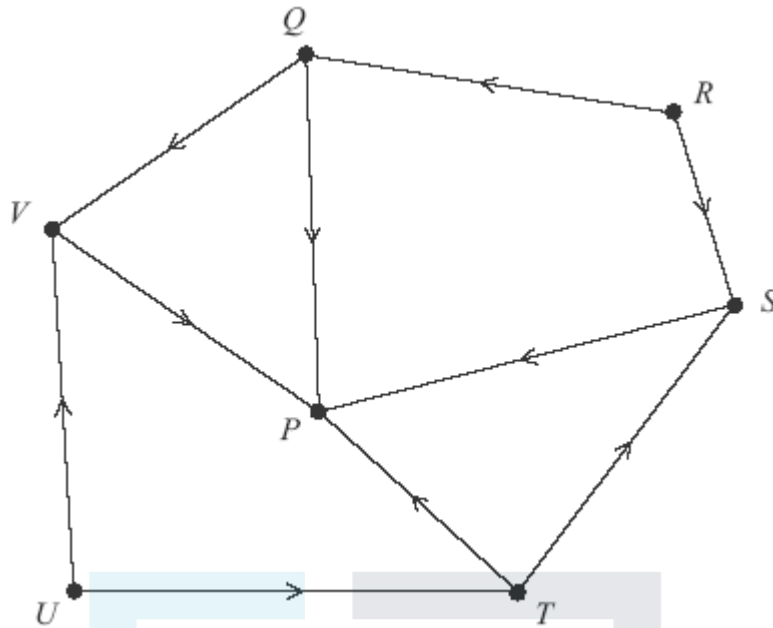
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(6)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2

Maximum flow is _____ passengers per minute



(2)

- (e) On a particular day, there is an obstruction allowing no more than 50 passengers per minute to pass through vertex V . State the maximum number of passengers that can move through the network per minute on this particular day.

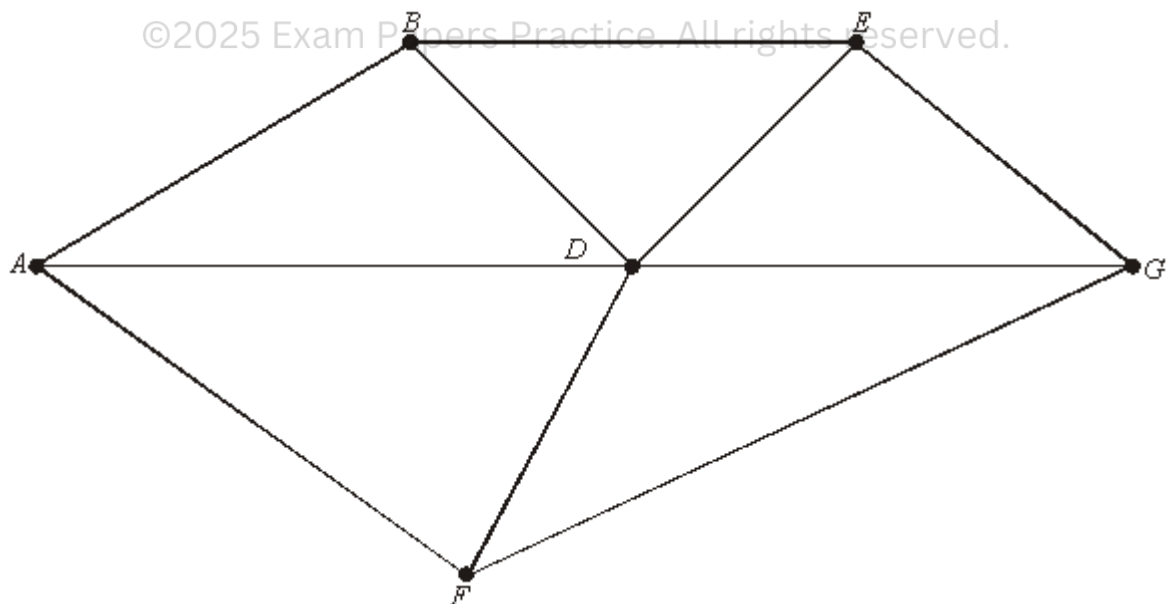
(2)

(Total 14 marks)

Q9.

[Figures 1 and 2, below, are provided for use in this question.]

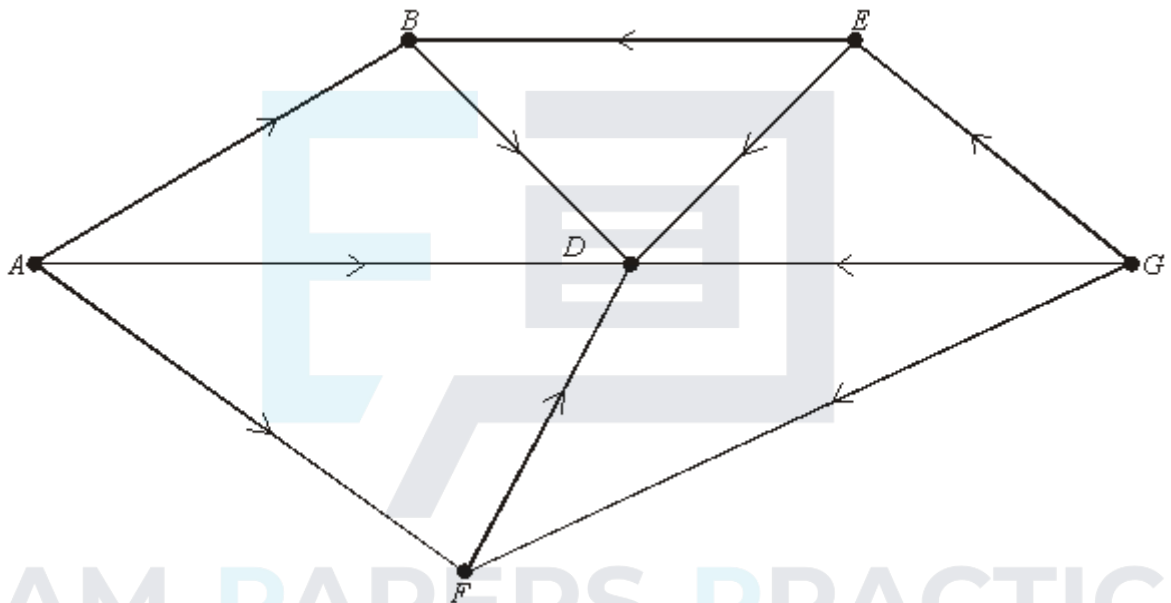
Figure 1



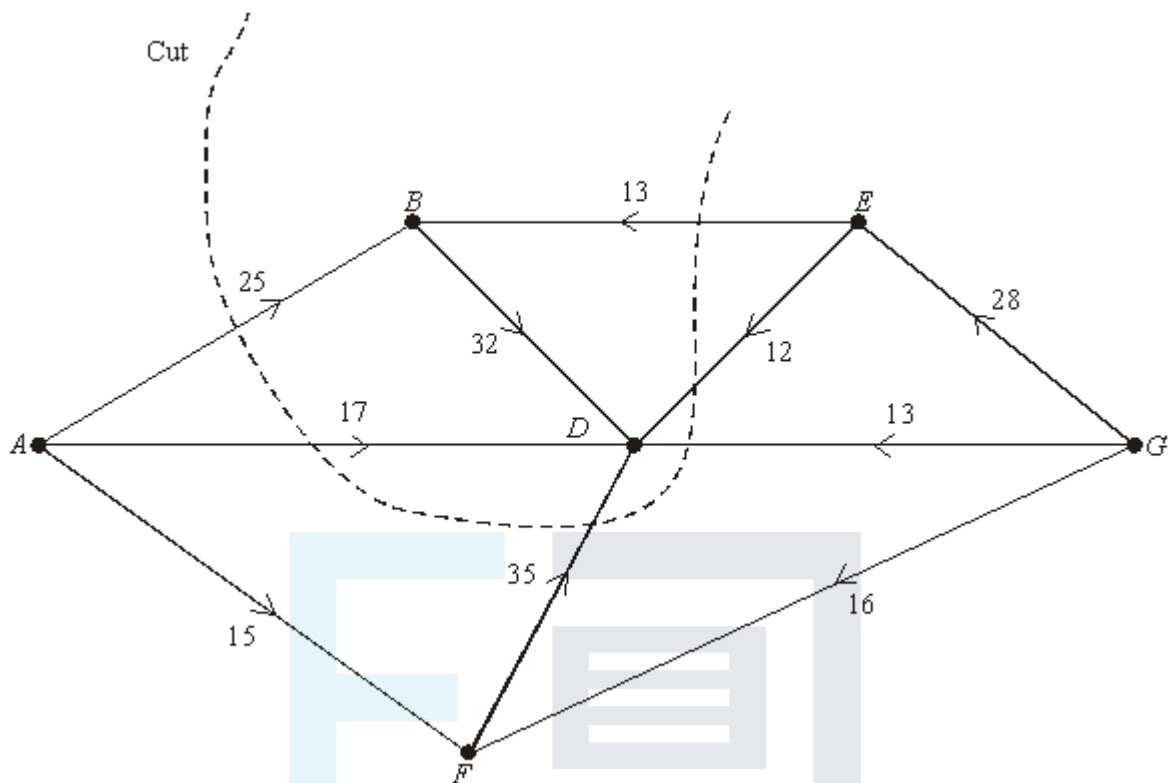
Route	Flow
ABC	

GED	
-------	--

Figure 2



The network shows the routes along corridors from the playgrounds A and G to the assembly hall in a school. The number on each edge represents the maximum number of pupils that can travel along the corridor in one minute.



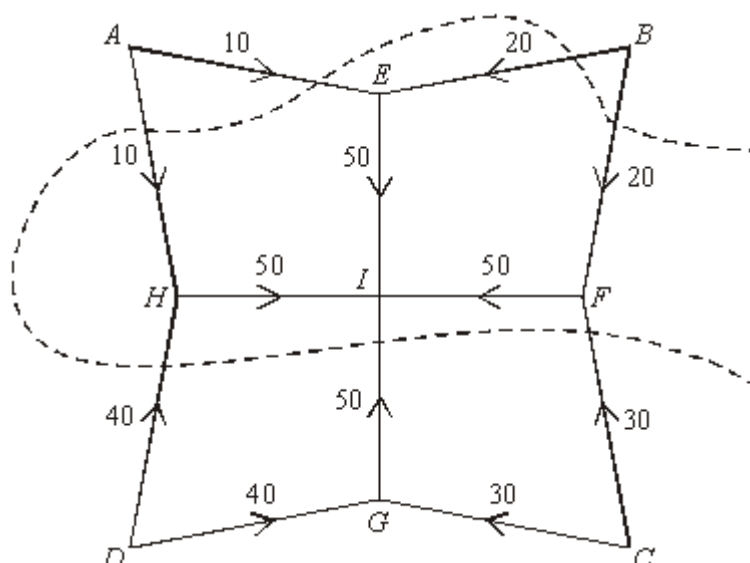
- (a) State the vertex that represents the assembly hall. (1)
- (b) Find the value of the cut shown on the diagram. (1)
- (c) State the maximum flow along the routes ABD and GED . (2)
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- (d) (i) Taking your answers to part (c) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow through the network. (6)
- (ii) State the value of the maximum flow and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow. (2)
- (iii) Verify that your flow is a maximum flow by finding a cut of the same value. (2)
- (e) On a particular day, there is an obstruction allowing no more than 15 pupils per minute to pass through vertex E . State the maximum number of pupils that can move through the network per minute on this particular day. (2)
- (Total 16 marks)

Q10.

The following diagram shows the maximum number of cars that can travel along each section of a one-way system in a 5 minute interval.

- (Total 7 marks)**

The following diagram represents a network of pipes used in a water system in a garden. The number on each arc represents the maximum volume of water, in cubic centimetres, that can flow along the corresponding pipe in one second.



- (1)

- (b) State the vertex that represents the sink. (1)
- (c) Find the value of the cut X . (2)
- (d) (i) On **Figure 1**, starting from a position of zero flow, use flow augmentation to find the maximum flow.

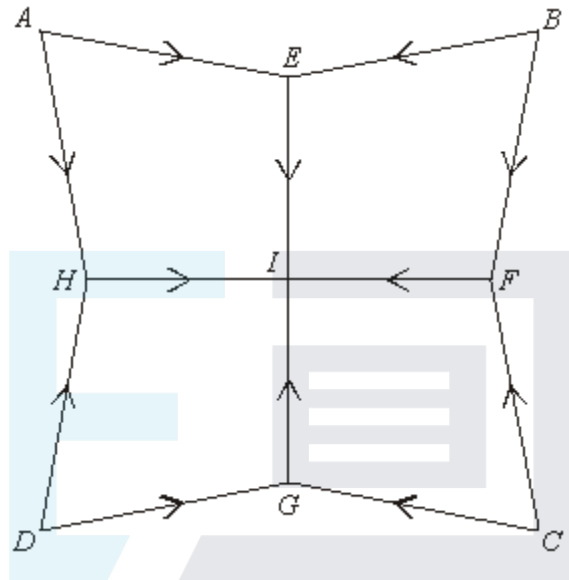


Figure 1

- (ii) State why this flow is maximum. (1)
- (e) On a particular day there is a restriction at vertex E which allows a maximum flow through E of 20.

On **Figure 2**, find, by inspection, the maximum flow through the network on this day.

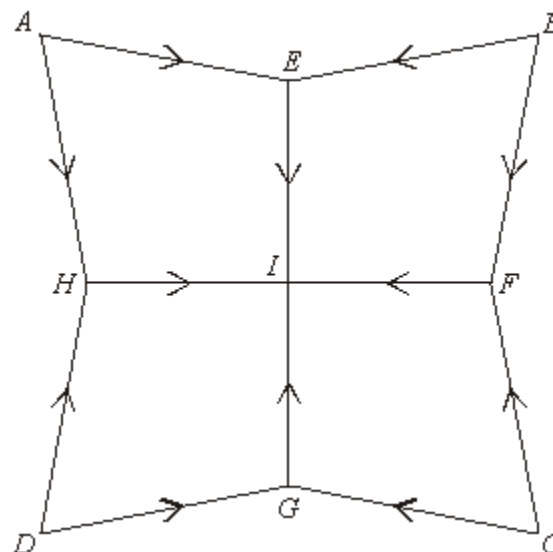
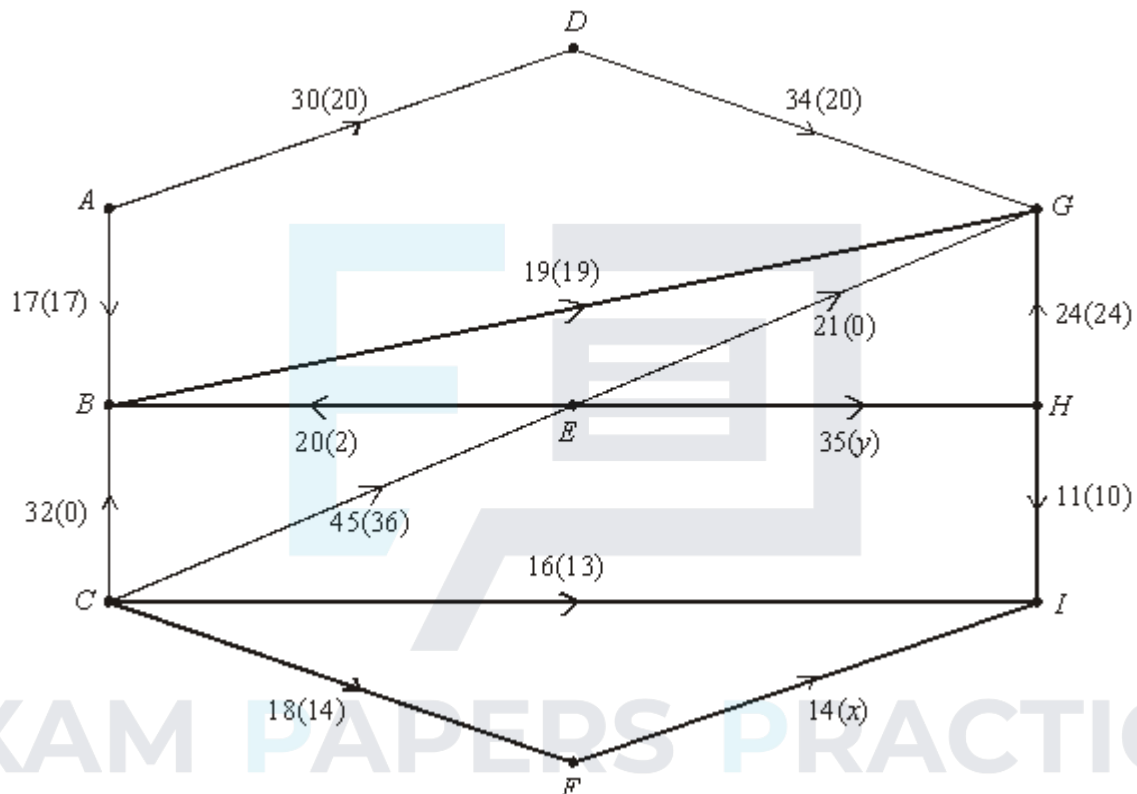


Figure 2

(2)
(Total 11 marks)

Q12.

The following diagram shows a capacitated directed network. The number on each arc is the capacity of that arc. The numbers in brackets represent an initial feasible flow through the network.

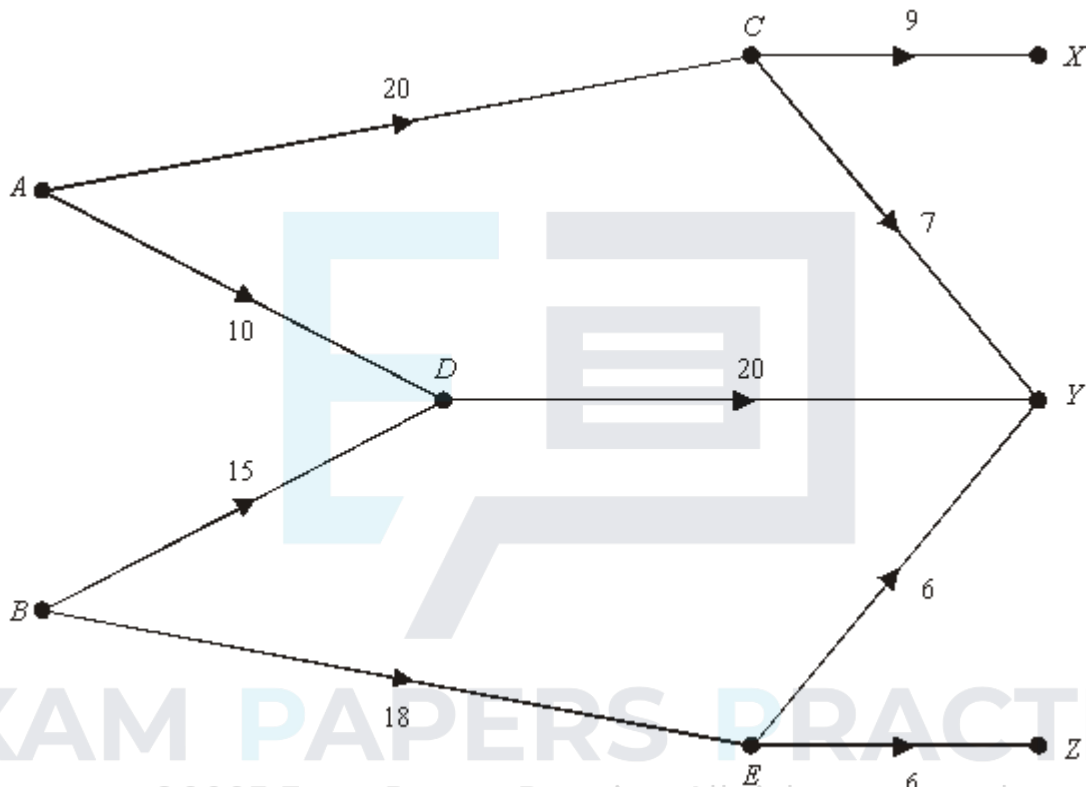


- (a) List:
- (i) the source vertices; (1)
 - (ii) the sink vertices. (1)
- (b) Find the value of:
- (i) the initial flow; (1)
 - (ii) x ; (1)
 - (iii) y . (1)
- (c) On the diagram above, starting from the given initial flow, use flow augmentation to find a flow of 122. (6)

(Total 11 marks)

Q13.

A greengrocer has two suppliers, A and B , and three storage depots, C , D and E . He needs to transport his stock to three retail outlets X , Y and Z . The capacities of the possible routes, in van loads per week, are shown in the following diagram.



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Figure 1

- (a) Add a super-source (S) and a super-sink (W) on **Figure 1** to obtain a single source, single sink capacitated network. Show the capacities of each arc you have added.

(2)

- (b) State the maximum flow along the routes $SADYW$ and $SBEZW$.

(2)

- (c)

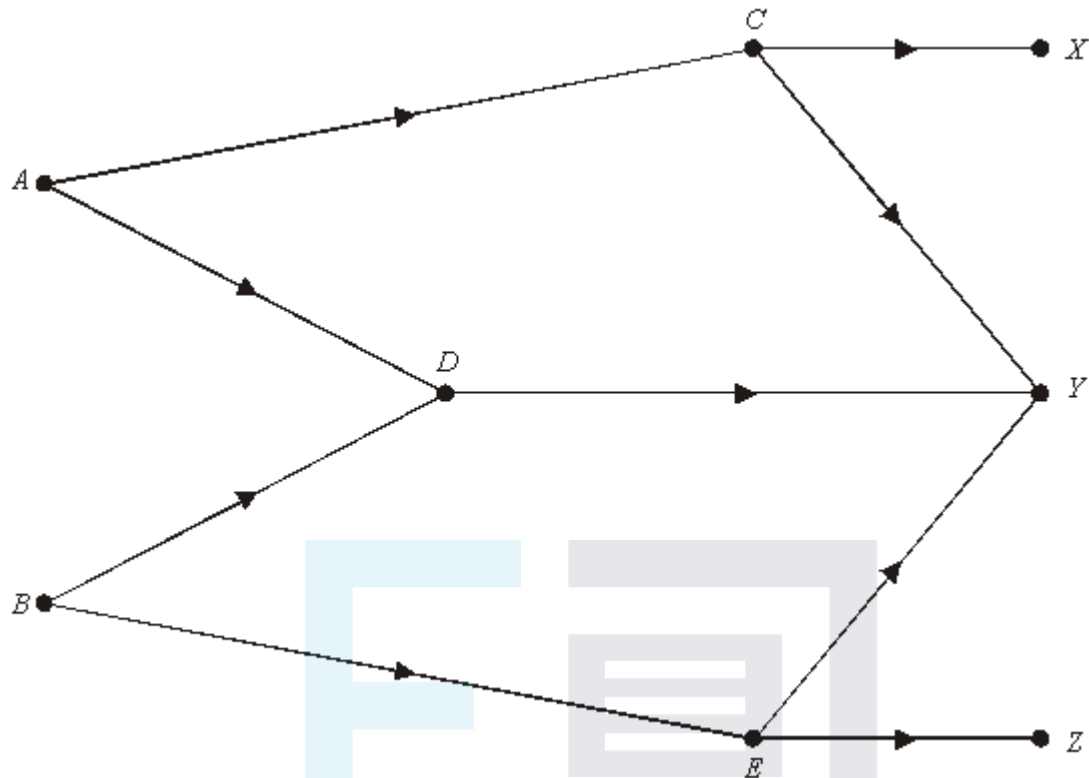


Figure 2

- (i) Show your answers to part (b) on **Figure 2** and, taking this as the initial flow pattern, use flow augmentation to find the maximum flow from S to W .

(6)

- (ii) Prove that your flow is maximal.

(2)

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Q14.

- (a) Draw a graph to represent the following matrix.

From \ To	A	B	C	D
A	0	2	1	0
B	1	0	1	1
C	1	1	2	2
D	1	0	3	1

(3)

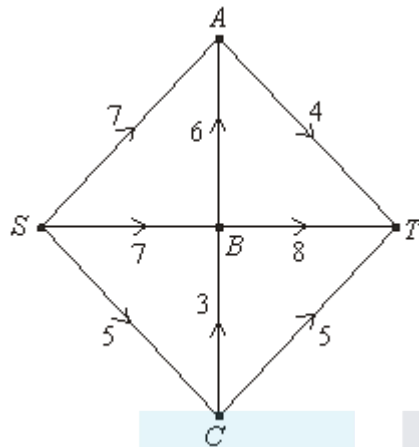
- (b) Explain from the matrix why your graph must be a directed graph.

(1)

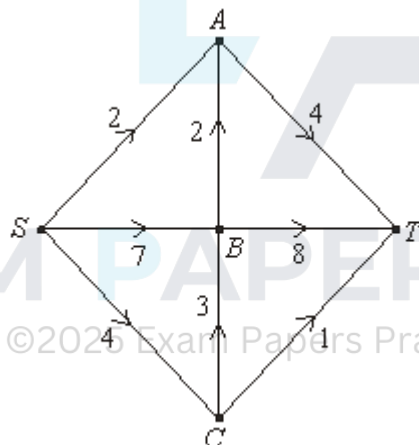
(Total 4 marks)

Q15.

The diagram shows the capacities of the arcs of a network.



- (a) Find a cut of 16 in the network above. (2)
- (b) The next diagram shows a flow of 13 from S to T in the same network.

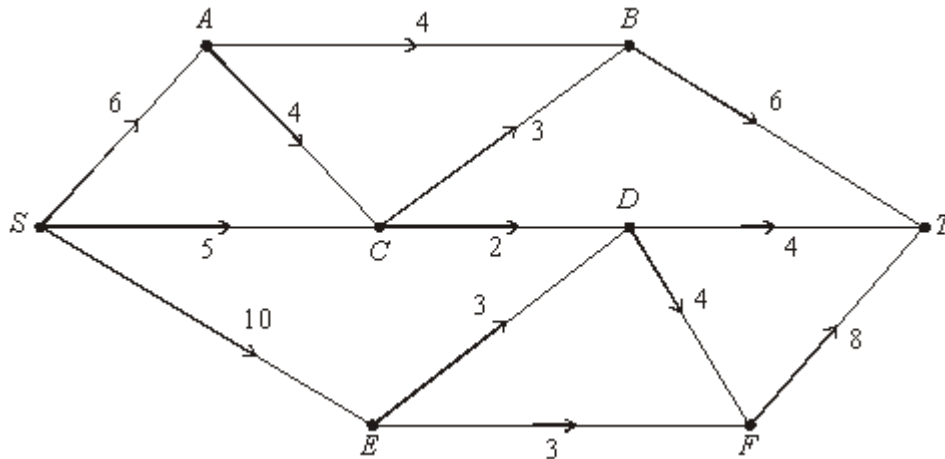


- (i) Find a flow-augmenting path from S to T which uses just one other vertex and which raises the given flow of 13 to a flow of 14. (2)
- (ii) Find a further flow-augmenting path from S to T which raises the flow from 14 to 16. (2)
- (c) How do you know that the flow of 16 is the maximum possible in this network? (1)
- (d) The capacity of one arc of the network is to be increased in order to enable a flow of 19 from S to T . State which arc should be chosen and give a reason. (3)

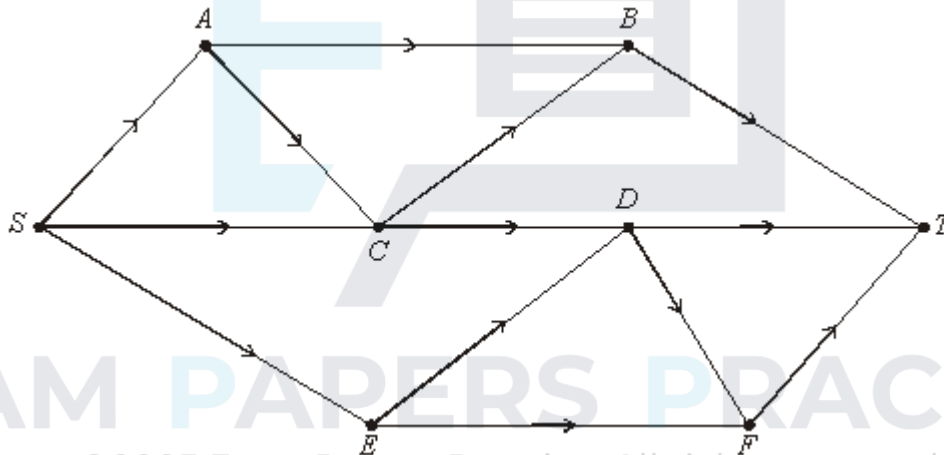
(Total 10 marks)

Q16.

The network shows a system of pipes and their capacities:



- (a) Starting from a zero flow, use flow-augmenting paths on the figure below to find a maximum flow from S to T .



- (b) Find a cut which confirms that your flow in (a) is a maximum. (5)
- (c) (i) Show that in any flow from S to T there can be at most a flow of 6 in the pipe SE . (2)
- (ii) Show that, if the pipe AC is saturated during a flow from S to T , there can be at most a flow of 1 in the pipe SC . (1)
- (iii) Deduce that in any maximum flow from S to T the pipe AC will not be saturated. (2)

(2)
(Total 12 marks)